Box 1: Metaecosystem multitrait-based biodiversity model

We outline a coevolutionary multitrait-based metaecosystem model to evaluate the effect of trait architecture (i.e., the covariation among biotic, abiotic and migration traits, the BAM traits) on biodiversity rescue (Figure 1). We explore two trait architectures rooted in empirical and theoretical studies (Armbruster et al. 2014): the modular and the trait integration architectures. The modular scenario takes into account independent traits, i.e. each trait contributes independently to fitness (Figure 1 left). The integration trait architecture incorporates the biotic trait as the main contributor to fitness with the abiotic and the migration traits depending on the evolution of the biotic trait (Fig 1, right).

Our model considers a landscape with a network of patches connected by migration events and inhabited by resource and consumer populations comprising several individuals each. Biotic, abiotic and migration trait values are initially sampled from normal distributions for each individual phenotype i represented as $\mathbf{z} = [z_b,\, z_a,\, z_m]$. Population dynamics of resource species i is driven by the trait- and temporal-dependent fitness functions following

$$\mathrm{W_R}(\mathbf{z_{B_i}})^t_x = 1/(1 + exp(-\gamma({\mathbf{z_{B_i}}}^t_x - \mathbf{\hat{z}}^t_{\mathbf{B_{forallj}}x})^2)),$$

$$\mathrm{W_R}(\mathbf{z_{A_i}})^t_x = exp[-\gamma({\mathbf{z_{A_i}}}^t_x - \mathbf{\hat{z}}^t_{\mathbf{A_i}x})^2],$$

$$\mathrm{W_R}(\mathbf{z_{M_i}})^t_x = exp[-\gamma({\mathbf{z_{M_i}}}^t_x - \hat{\theta}_{\mathbf{D}x})^2],$$

and for the fitness functions for the consumers are the following

$$\mathrm{W_C}(\mathbf{z_{B_i}})^t_x = exp[-\gamma({\mathbf{z_{B_i}}}^t_x - \mathbf{\hat{z}}^t_{\mathbf{B_{forallj}}x})^2],$$

$$\mathrm{W_C}(\mathbf{z_{A_i}})^t_x = exp[-\gamma({\mathbf{z_{A_i}}}^t_x - \mathbf{\hat{z}}^t_{\mathbf{A_i}x})^2],$$

$$\mathrm{W_C}(\mathbf{z_{M_i}})^t_x = exp[-\gamma({\mathbf{z_{M_i}}}^t_x - \hat{\theta}_{\mathbf{D}x})^2],$$

where ${\mathbf{z_T}}^t_x$ is the trait value T of phenotype i at time t and site x, θ_x is the multivariate fitness optimum, ω is the covariance matrix (Fig 1) (Lande, 1980; Melo and Marroig, 2014), and γ determines the interaction sensitivity to deviations from the biotic (i.e., coevolutionary selection strength), abiotic and migration optimum.

If the covariance matrix, ω, is diagonal, representing our modular scenario (Fig. 1, left), then there is no correlated stabilizing selection. If the covariance matrix is not diagonal and considering our magic scenario (Fig. 1, right), then there is correlated stabilizing selection with the biotic trait determining the selection on the other traits. The biotic optimum changes in time and we define it as the minimum and maximum mean trait distance across all populations in site x for the consumer and the resource biotic trait value, respectively (eqs. 1 and 4, respectively). The dispersal optimum is given by the mean distance in the landscape (i.e., each site has coordinates from which we calculate the Euclidian distance between each pair of sites). Phenotypes with migration trait value near to this optimum maximize fitness for this trait only for the modular but not for the magic trait scenario. The abiotic optimum is taken initially as the mean of the distribution and it remains the same during the dynamics.

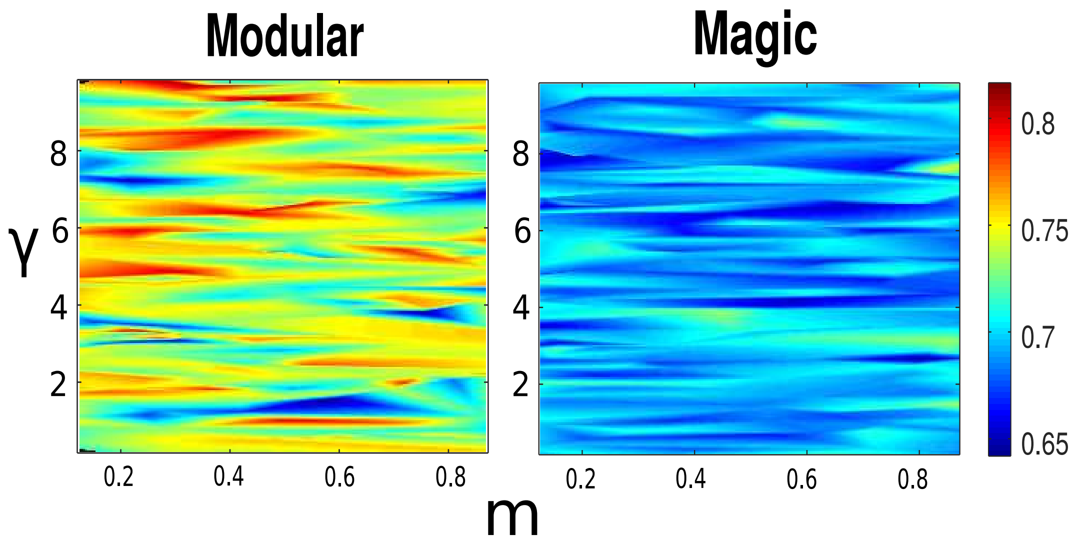

Figure 1: **Modular (left) and Magic (right) scenarios**: To evaluate the importance of coevolutionary selection and migration as rescue mechanisms in modulating biodiversity dynamics, we explore the formation of hot and cold spots of species persistence in the landscape along a gradient of migration and coevolutionary selection strength. **Left**. The modular scenario predicts higher rescue than the magic scenario across the whole phase space with hot spots occurring for low-to-medium migration and medium-to-high coevolutionary selection. **Right**: The magic scenario predicts generally low-to-moderate- rescue. The highest rescue values are in the low coevolutionary selection range along a broad range of migration.

At each time step a certain percentage of individuals are selected randomly to die and are replaced by birth or migration. The time dependent fitness function in site x, $\mathrm{W}(\mathbf{z})_x^t$, determines the mother of the offspring. To obtain the phenotypic value for the offspring, $z_o = [z_{ob}, z_{oa}, z_{od}]$, we add to each trait value of the mother the environmental deviation, $\mathbf{e}$, taken from a multivariate uniform distribution with range ± 0.001 in all dimensions (i.e., independent sampled values for the modular but correlated for the integration trait scenario). We run our model for 100 replicates with 50 generations each for the modular and integration scenarios. To evaluate the importance of coevolution and migration as rescue mechanisms in modulating biodiversity dynamics, we explore how the formation of hot and cold spots of species persistence in local and regional biodiversity changes in a gradient of migration and coevolutionary selection strength. Codes and outputs can be downloaded from https://github.com/melian009/Cometa).

1. Bibliography

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